

Impact of a Brief Guided Meditation Intervention on Perceived Stress and Anxiety

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Abstract

- Stress is a major public health concern in modern society (Piao et al., 2024).
- Acute stress and anxiety negatively impact both mental and physical health and can interfere with daily functioning and overall well-being (Schneiderman et al., 2005).
- Mindfulness training, including practices like guided meditation, has been shown to help with various life challenges, such as reducing symptoms of depression (Schreiner & Malcolm, 2008).
- This study examined whether a brief ten-minute guided meditation session could reduce perceived stress and anxiety in a local community setting.**
- Participants attended a short guided meditation session conducted in community-based environments and completed anonymous post-session surveys.
- Results indicated a reduction in perceived stress and anxiety following the meditation session.** Most participants reported moderate to high stress levels prior to the session and felt calmer and more relaxed afterward.
- Findings suggest that brief guided meditation sessions may reduce stress and anxiety and support emotional well-being.

Methods

- Surveys were conducted at a senior living home (n = 37), a public library (n = 65), and a high school faculty group (n = 162), AP Psychology classes (n= 177), classes at Del Norte High School (n= 53), online classes (n= 54), for a total of 548 participants across a wide age range.
- Participation was voluntary, and responses were collected anonymously to reduce social desirability bias.
- Participants completed pre- and post-session anxiety/stress ratings using a standardized one-to-five Likert scale before and after a ten-minute guided meditation session.
- Survey questions were neutrally worded and consistently formatted to minimize response bias.
- Consistent procedures, standardized measures, and anonymity helped reduce measurement error and bias.
- Survey data were analyzed using basic percentages in Google Excel to compare pre- and post-session anxiety/stress levels.
- Bar graphs and pie charts illustrated trends in anxiety/stress reduction.
- Key limitations included self-reported data, differences in prior meditation experience, and environmental distractions in community settings (ambient noise, movement of others, limited privacy).

Results

Age Demographics of Survey Sample

25-44 Under 12 65+ 45-64 12-17 18-24

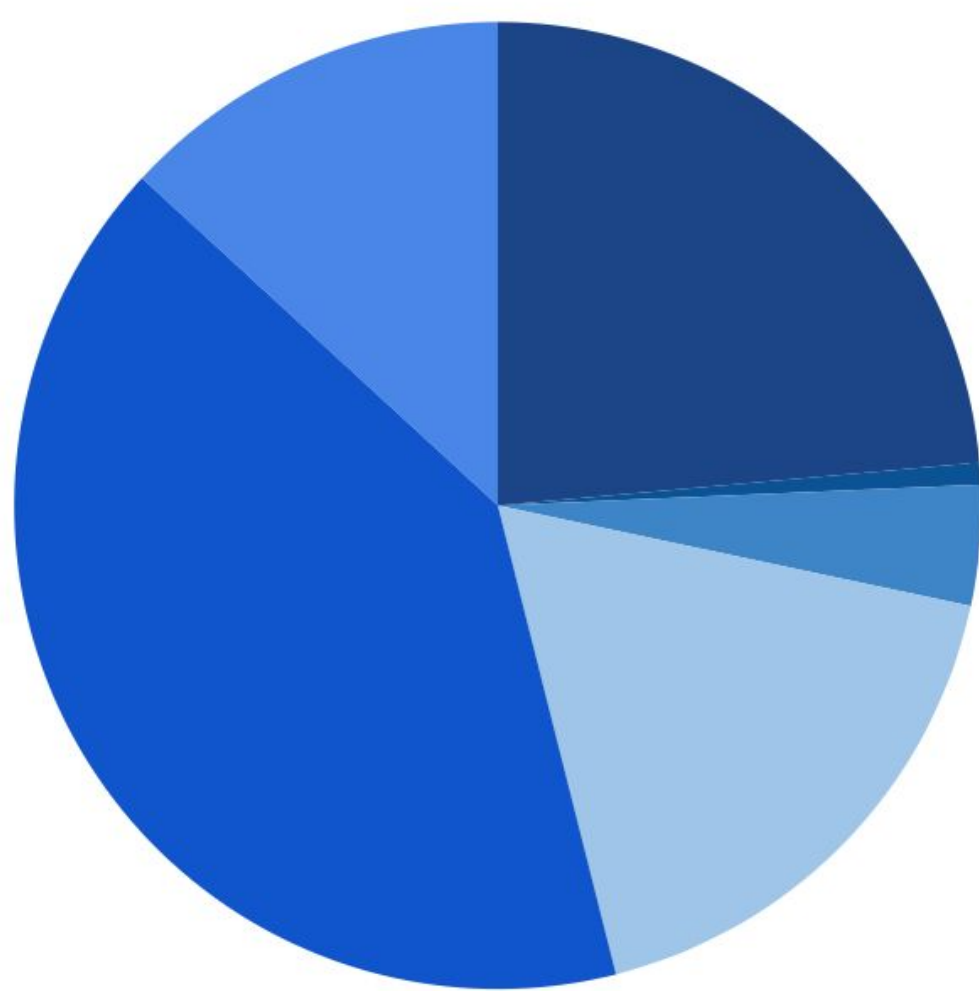


Figure 1. Pie chart showing participant demographic characteristics.

Figure 1 displays the age distribution of the 66 participants. Participants most frequently fell within the 45–64 (33.3%) and 25–44 (28.8%) age groups, while fewer participants fell under ages 12 and 18–24. This distribution shows that the sample included primarily adults with representation across multiple age groups.

Distribution of Self-Reported Anxiety/Stress Levels Before the Session

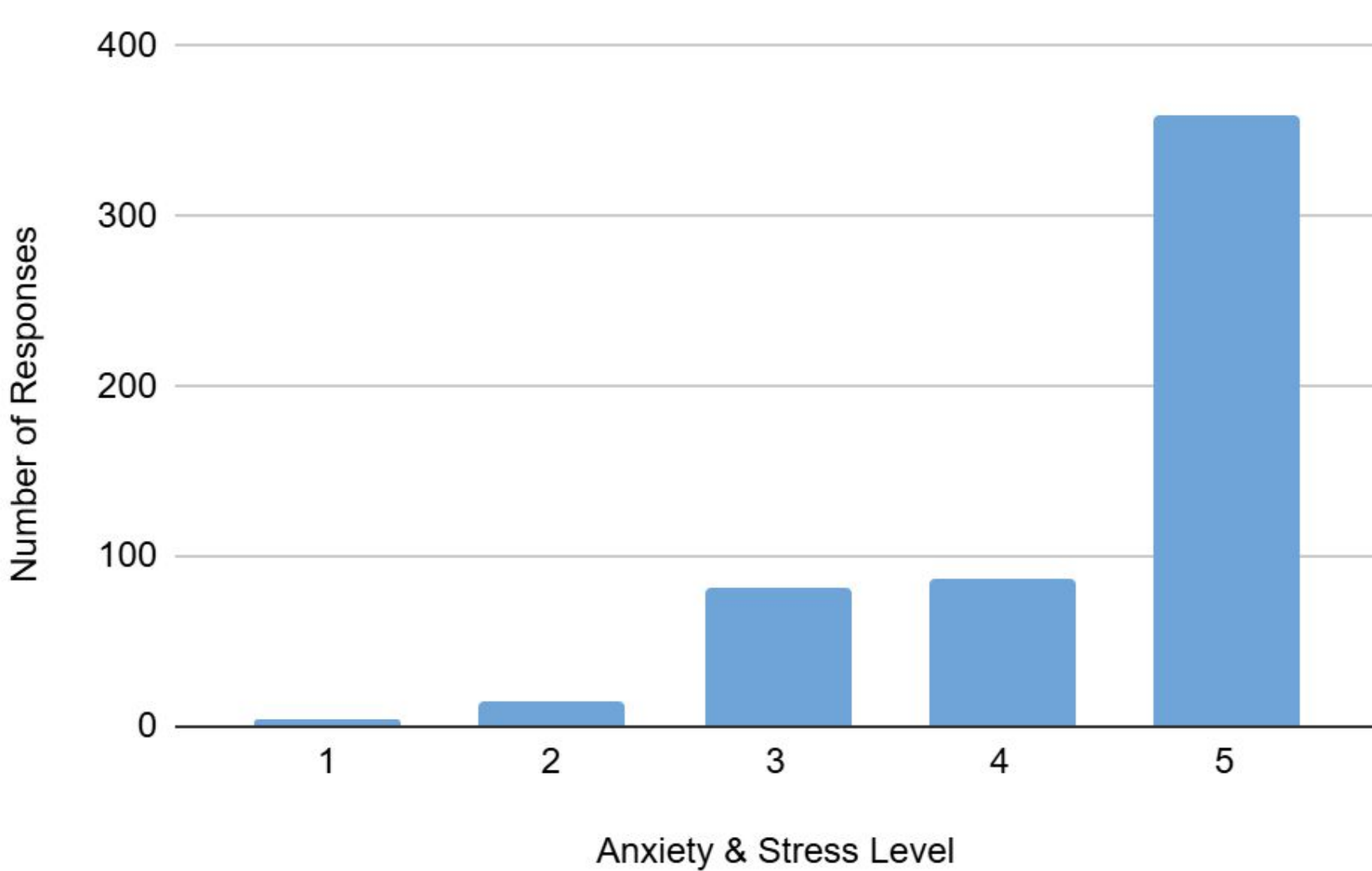


Figure 2. Bar graph displaying self-reported stress and anxiety levels prior to the guided meditation session.

Distribution of Self-Reported Anxiety/Stress Levels After the Session

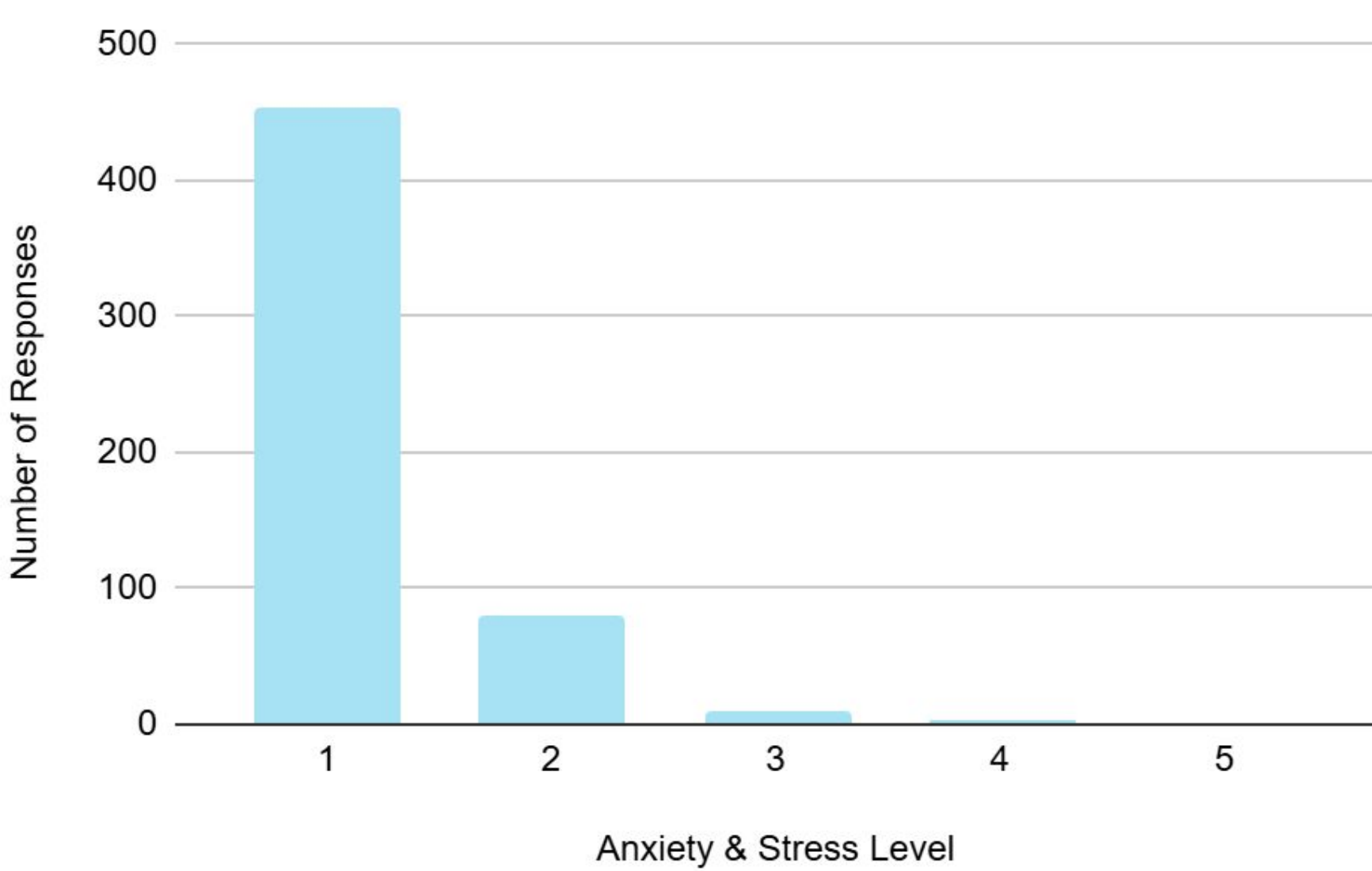


Figure 3. Bar graph displaying self-reported stress and anxiety levels following the guided meditation session.

Figures 2 and 3 compare anxiety/stress levels before and after the meditation session. Before the session, most participants reported moderate to high anxiety/stress (levels 4–5). After the session, responses shifted strongly toward the lowest levels (1–2), and very few participants reported moderate or high anxiety/stress. The paired-samples *t*-test showed a statistically significant decrease in anxiety/stress following the session, $t(546) = 49.54, p < .001$. Overall, participants showcased a **70.2% reduction** in mean anxiety/stress scores after a single 10-minute guided meditation, decreasing from $M = 4.20$ ($SD = 1.13$) pre-session to $M = 1.25$ ($SD = 0.56$) post-session.

Participant Responses to Post-Meditation Calmness

Yes, a lot A little Not Really

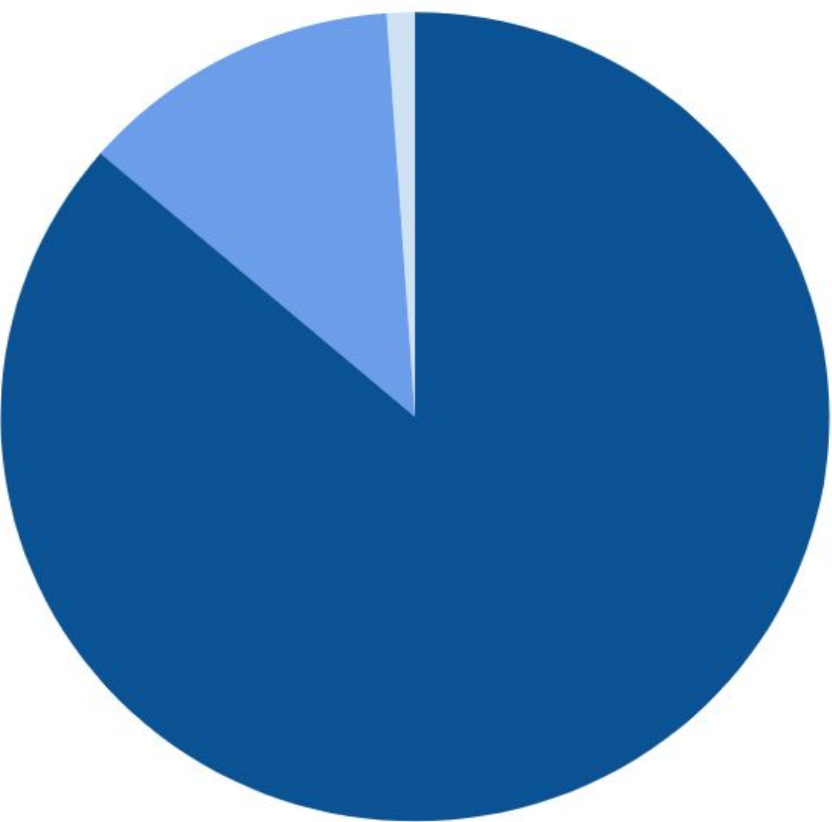


Figure 4. Pie chart summarizing participant responses to the guided meditation experience.

Figure 4 shows that most participants reported feeling calmer following the guided meditation session. Approximately **85%** of participants reported feeling **significantly calmer**, about **13%** reported **feeling a little calmer**, and approximately **2%** reported **no noticeable improvement**. No participants reported feeling more stressed or anxious after the session. This shift toward greater calmness demonstrates the immediate positive effect of the 10-minute guided meditation across participants.

Conclusions

- Data from **547 participants** were analyzed using descriptive statistics and pre-/post-session comparisons.
- Before meditation, most participants reported **moderate to high anxiety/stress**, with the highest frequencies at **levels 4 and 5** on the 1–5 scale.
- After the 10-minute guided meditation, anxiety/stress ratings shifted strongly toward **levels 1 and 2**, and virtually no participants reported high anxiety/stress.
- A paired-samples *t*-test confirmed that this reduction in anxiety/stress was statistically significant ($t(546) = 49.54, p < .001$).
- Approximately **85%** of participants stated they felt significantly calmer after the session, supporting the numerical decrease in anxiety/stress ratings.
- Short mindfulness interventions produce immediate psychological benefits across participants.
- Meditation requires no cost, little time, and no special equipment, allowing use in schools, community programs, and at home.
- These findings suggest that brief mindfulness practices can serve as an accessible first step in managing stress and anxiety.**

References

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